

R Notebook

Dependencies

```
install.packages("lubridate")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'  
## (as 'lib' is unspecified)
```

```
install.packages("tidyverse")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'  
## (as 'lib' is unspecified)
```

```
install.packages("dplyr")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'  
## (as 'lib' is unspecified)
```

```
install.packages("viridis")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'  
## (as 'lib' is unspecified)
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.6  
## v forcats    1.0.1      v stringr    1.6.0  
## v ggplot2    4.0.1      v tibble     3.3.0  
## v lubridate  1.9.4      v tidyr      1.3.2  
## v purrr      1.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
```

```
library(dplyr)
```

```
library(ggplot2)
```

```
library(viridis)
```

```
## Loading required package: viridisLite
```

Loading data

```
# Load data
```

```
ipfv_df<-read.csv("Intimate Partner and Family Violence.csv", stringsAsFactors = FALSE)  
head(ipfv_df)
```

```
##   X_id INDEX_ REPORT_YEAR REPORT_MONTH REPORT_DOW   HISTORICAL  
## 1     1     1         2024         July      Monday Not Historical  
## 2     2     2         2024    February      Tuesday Not Historical  
## 3     3     3         2024    February      Tuesday Not Historical
```

```
## 4      4      4      2024      February      Tuesday Not Historical
## 5      5      5      2024      February      Tuesday Not Historical
## 6      6      6      2024      February      Tuesday Not Historical
##      FAMILY_VIOLENCE_FLAG      FAMILY_VIOLENCE_RELATION DIVISION PREMISES_TYPE
## 1              Intimate Former Intimate Relationship      D33      Outside
## 2              Intimate Former Intimate Relationship      D14      Other
## 3              Intimate      Intimate Relationship      D32      House
## 4              Family Other Familial Relationship      D32      Apartment
## 5              Intimate      Intimate Relationship      D53      Apartment
## 6              Intimate      Intimate Relationship      D11      Apartment
##      HOOD_158      NEIGHBOURHOOD_158 HOOD_140      NEIGHBOURHOOD_140 COUNT_
## 1      053      Henry Farm (53)      053      Henry Farm (53)      1
## 2      084      Little Portugal (84)      084      Little Portugal (84)      1
## 3      031      Yorkdale-Glen Park (31)      031      Yorkdale-Glen Park (31)      1
## 4      031      Yorkdale-Glen Park (31)      031      Yorkdale-Glen Park (31)      1
## 5      098      Rosedale-Moore Park (98)      098      Rosedale-Moore Park (98)      1
## 6      085      South Parkdale (85)      085      South Parkdale (85)      1
```

```
names(ipfv_df)
```

```
## [1] "X_id"      "INDEX_"
## [3] "REPORT_YEAR"      "REPORT_MONTH"
## [5] "REPORT_DOW"      "HISTORICAL"
## [7] "FAMILY_VIOLENCE_FLAG"      "FAMILY_VIOLENCE_RELATION"
## [9] "DIVISION"      "PREMISES_TYPE"
## [11] "HOOD_158"      "NEIGHBOURHOOD_158"
## [13] "HOOD_140"      "NEIGHBOURHOOD_140"
## [15] "COUNT_"
```

```
class(ipfv_df)
```

```
## [1] "data.frame"
```

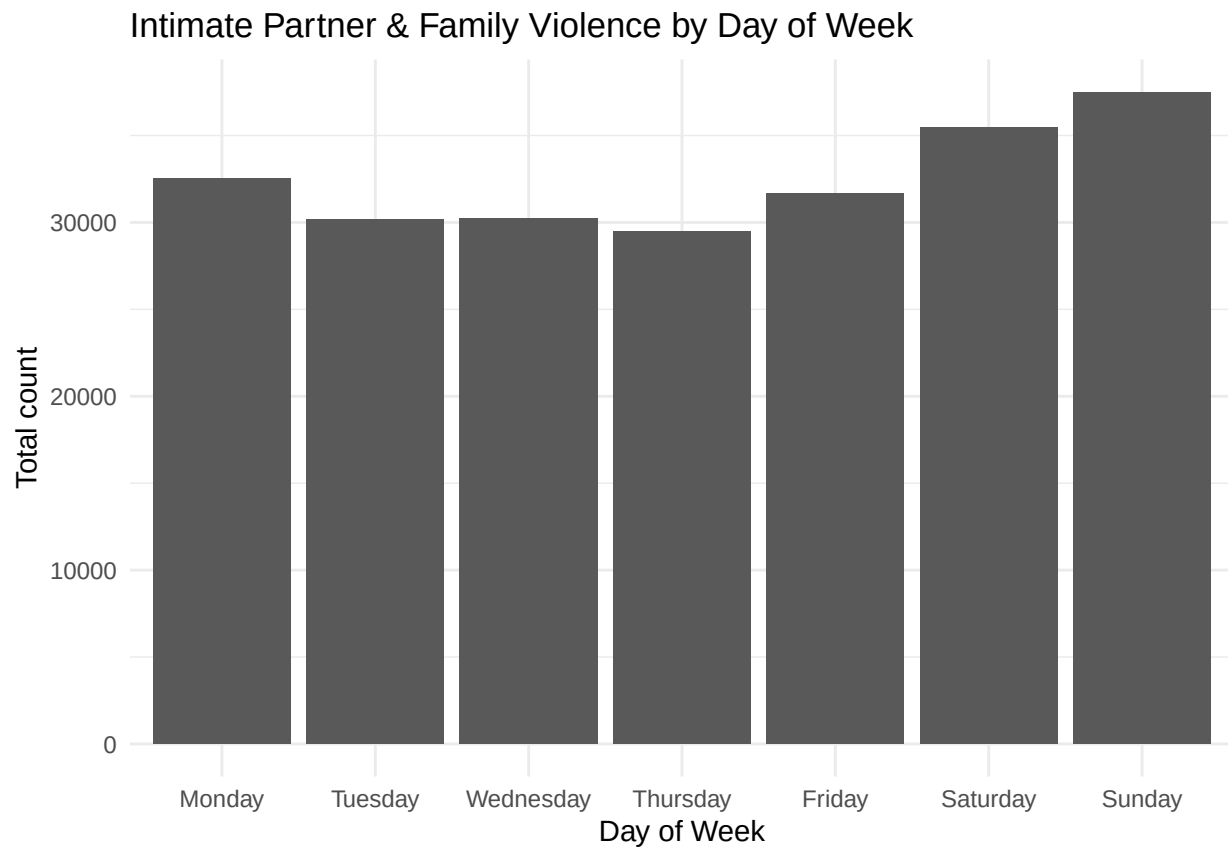
First visual count by day of the week

```
library(dplyr)
library(ggplot2)

#bar plot displaying total count by day of the week
ipfv_df$REPORT_DOW <- factor(
  ipfv_df$REPORT_DOW,
  levels = c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday")
)

dow_counts <- ipfv_df %>%
  group_by(REPORT_DOW) %>%
  summarise(total = sum(COUNT_, na.rm = TRUE), .groups = "drop")

ggplot(dow_counts, aes(x = REPORT_DOW, y = total)) +
  geom_col() +
  labs(
    title = "Intimate Partner & Family Violence by Day of Week",
    x = "Day of Week",
    y = "Total count"
  ) +
  theme_minimal()
```



Second visual count by relationship

```
#bar plot displaying total count by relationship
relation_counts <- ipfv_df %>%
  group_by(FAMILY_VIOLENCE_RELATION) %>%
  summarise(
    total = sum(COUNT_, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(total))

ggplot(relation_counts,
  aes(x = reorder(FAMILY_VIOLENCE_RELATION, total),
    y = total)) +

  geom_col(fill = "steelblue") +

  coord_flip() + # easier to read long labels

  labs(
    title = "Intimate Partner & Family Violence by Relation",
    x = "Relation Type",
    y = "Total count"
  ) +

  theme_minimal()
```

